Secure Campus Messaging at Scale: An Open-Source, Commodity-Based Edge Design

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A. Proposal Overview

A.1 Problem Summary

K-12 schools rely on reliable information and are tasked with distributing it to their stakeholders (students, educators, staff members, parents, caretakers, and other community members). They use many methods of electronic communication to deliver information to their stakeholders, including email lists, social media groups, newsletter platforms, 3rd-party apps, and more. It is paramount that communications from trusted platforms are available, as they provide important and sometimes crucial information for students and caregivers to manage their students' educational pathways and experiences. If each student is to do their best in their educational experience, they need the opportunity to participate in every program offered by the institution that fits their needs and interests.

Also, K-12 schools commonly lack access to modern technology and solutions. These environments are often over-tasked with waves of activities to fulfill a school-year calendar, and these activities are spread across various waves, leaving many education technology concerns unaddressed when the time to develop a solution and/or the budget to complete a project is available. In a well-researched, published report titled "Challenges and solutions when using technologies in the classroom."[1]Researchers from Arizona State University found that schools do not get access to education technology solutions due (in part) to three primary constraints: access to equipment and implementation, training on how to use the technology, and support barriers to technology integration (Johnson et al., 2016). Solutions that are not cost-effective, easy to deploy and manage, and do not fit the K-12 environment often leave gaps in the technology a school needs to operate effectively.

To support this concept, this paper proposes the design of an electronic messaging system to deliver secure displayboard functionality using well-known open-source packages and protocols, paired with low-cost commodity hardware, to create a reliable, secure system for use in the K-12 school context. As reflected in "Free and Open Source Software in Education", the author describes why open-source technology is effective in education contexts and states: "... a significant advantage to FOSS educational software is that it typically runs on less-powerful computer hardware than proprietary software, which reduces or even eliminates the need for cash-strapped school districts to upgrade their existing

[1] Johnson, A. M., Jacovina, M. E., Russell, D. G., & Soto, C. M. (2016). Challenges and solutions when using technologies in the classroom. In S. A. Crossley & D. S. McNamara (Eds.), Adaptive educational technologies for literacy instruction (pp. 13–29). Taylor & Francis. https://files.eric.ed.gov/fulltext/ED577147.pdf

equipment so that those resources can be spent elsewhere."[2] The design system supports the best of open technology, paired with commodity hardware (likely to be found in K-12 settings), to deliver commercial-grade displayboard technology at very low cost, so resources can be used elsewhere. With this solution, more schools would have access to better communication technology, presenting more opportunities for students and increasing overall participation in their educational pathways. The paper will analyze the use cases and problems that an open-source, commodity-based secure messaging platform can deliver, and detail a working, lightweight prototype to complete the overall capstone effort.

A.2 IT Solution

To address this problem, this project will develop an IT solution using widely available open-source technologies and consumer-grade, low-cost hardware. As K-12 educational agencies are typically tasked with many responsibilities and have many needs, low-cost IT solutions can effectively fill gaps where other commercial solutions may not be affordable. The IT Solution provided must fit the agency's needs of the school system: it needs to be secure and unhackable from standard methods, it needs to provide reliable and relevant information for its stakeholders, and

This solution will provide an educational agency, such as a school or district, with a workable messaging delivery system featuring rotating displays that use low-cost software and hardware and commonly available school interfaces and standards to operate on. Schools and agencies often have high-resolution displays in the form of large-format televisions and projectors with HDMI ports. This solution will support working with many school display devices in their asset inventory. Schools and agencies commonly use wireless networking with common WiFi standards; this solution will support those standards. These organizations commonly have Raspberry Pi computing devices available, which are low-cost educational devices. This solution will support those devices in the proposal and prototype solution. (The next step, if taken to complete production on the concept, might be to support low-cost laptop devices such as Chromebooks and PC devices; however, this is not a matter for investigation in this proposal. The solution should be 85% ready for other computing devices.)

This solution will also use a low-cost information source for distribution that a school district or agency is likely to use in its context or setting. We would assume an IT administrator would use the solution, someone in that role, or a small team of administrators responsible for setting the messages that rotate on the display targets. As we build this solution in the next phase, the author will evaluate several low-cost strategies to distribute information securely, without overburdening the project with excessive build or expense, nor burdening the end-user with a complicated management mechanism. For the data entry system to work in this solution, its requirements are:

[2] Donnelly, M. (2021). Free and open source software in education. EBSCO Research Starters. https://www.ebsco.com/research-starters/computer-science/free-and-open-source-software-education

- The solution must be straightforward and a reasonable office data management task for an IT administrator or office staff member to use with a reasonable instruction set and/or demonstration. They should be able to understand the data entry system within 10-15 minutes after reading 1-3 pages of information or watching a 5-10 minute video. It should be on the level of other K-12 educational agency administration tasks - such as attendance management, building roster management, building operations and ticketing management, and other day-to-day tasks found within a school-like environment. The solution must not assume advanced or expert-level skills to operate.
- The solution must be secure from end to end, reliable, and, if the system fails, provide mechanisms for administrative users to understand its state of operation. They must be able to explain to the agency what happened to the system and when it is expected to be back online, using information from the solution. The solution should take measures to prevent hacking from various threat groups, as defined by standard models, such as NIST Special Publication 800-53[3] Moreover, SOC 2 guidelines, which education organizations are typically subject to under state and federal frameworks/are used to run their day-to-day operations with a high level of assurance. The information processed here will be general announcements. They will not contain sensitive or personally-identifiable information (PII) to warrant a high level of information security; however, we will secure this to a Moderate level as defined by NIST standards, or a public-trust level to ensure the system operates in a manner found in most public institutions (like airports, courthouses, event centers, and so on).
- The solution should be modern and present modern-day approaches to today's IT challenges. For example, techniques such as two-factor authentication (2FA) should be considered and implemented in the solution to ensure real-time verification using standard devices found in a school environment, thereby meeting the security needs listed above. 2FA is commonly used in many K-12 solutions today, where a user is asked to enter an additional code from a text message or work-assigned email address, providing a factor that few other people would have access to (or would be very expensive to replicate if not for that end-user). The solution should use modern data transfer and secure storage techniques - such as using today's secure web transfer standards like SSL/TLS, HTTPS, REST APIs, and other modern protocols- and through the final solution prototype, will have a post-execution write-up detailing these methods for the final end-to-end solution. The solution will use standard open-source encryption techniques that support low/no-cost goals and also provide commercial & government-grade protection for information during delivery ("in transit") and while stored ("at rest"). 2-way encryption protocols, such as AES, will be considered the final solution for "at rest" and "in transit" pipelines, which will utilize SSL/TLS to ensure information is not transferred outside a secure channel from end to end. Lastly, digital signatures, using open-source technology and NIST guidelines, will be utilized in this solution to ensure that messages generated have been securely

[3] National Institute of Standards and Technology. (2020). *Security and privacy controls for information systems and organizations* (NIST Special Publication 800-53; Rev. 5). U.S. Department of Commerce. https://doi.org/10.6028/NIST.SP.800-53r5

delivered and not tampered with during the message delivery pipeline. In combination, these techniques will protect from various threat actors - those unknown and at random from the outside, from various capabilities levels (from casual to organized hacker groups), and those inside the organization which may thwart for various reasons (from disgruntled students and staff members), to ensure the system is as reliable as possible to serve its purposes and agency needs.

With a secure, low-cost solution proposed and investigated in this capstone, a fictional school or district would have a reasonable, usable solution for displaying and rotating bulletin-type information, and would be able to do so in a secure way using software and hardware assets they are likely to have on hand. The proposal ties together many aspects learned in this program, such as secure information design, using various frameworks to ensure compliance in various controlled environments (here FERPA and federal & state regulations as they apply to K-12 education, however other standards like HIPPA for privacy adherence and PCI for sensitive information apply in concept as well), securing information at-rest and in-transit and various methods for doing so, how to evaluate overall solutions and more to tie many aspects together as a working prototype example. It should prove out that today's open-source and commodity-level technology can perform commercial-level operations with the right design, construction, orchestration, and applications; the results will be telling for what is possible in a realistic, real-world modeling environment.

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A.3 Implementation Plan

The implementation plan describes an approach for delivering the overall solution over 2 to 3 weeks. This plan will emulate a real-world project, but in an accelerated fashion as a proof-of-concept or prototype, to demonstrate the concept for capstone purposes and to fulfill its requirements for overall program completion. The author has a deep and extensive background in information technology, software development, program implementation and execution, technology startup execution, and project monitoring and delivery; so the best of the author's experience will be utilized here in implementation; combined with the many experiences and knowledge learned through the WGU B.S.C.S program with cybersecurity focus - including deepening in security protocols via CompTIA Security+, isolated network design and analysis via CompTIA Network+, open-source utilization from Linux certification, real-world analysis and application found in courses such as via Managing Information Security – C843, Digital Forensics in Cybersecurity – D431 and Legal Issues in Information Security – C841. This implementation plan is minimally viable and will pass common security reviews found in education agencies for information processing and storage at this level, for a moderate and/or public-trust type solution, as defined by its scope. Out-of-scope implications would be: operationalizing this at scale, commercial cost implications, the cost of actual security scans if a customer needed to use them, and other full go-to-market considerations. However, the proposed solution will provide a real-world, working concept to demonstrate for the first stage of the intended development.

The implementation plan will use the Scrum Alliance[4] "agile framework", which is a commonly adopted planning, execution, and management framework used in today's software development environments. This framework allows for efficient delivery of software using proven techniques and is well-adapted for new projects or projects with many unknowns. The Scrum Alliance agile framework is also well-suited for projects that are tolerant of or expect "scope changes" or "scope creep," where unknowns arise during project execution. It allows for decision changes or adjustments along the way and still enables the project to finish and deliver. Some project methodology frameworks, such as a Waterfall methodology, would bring more rigor to a guaranteed delivery for an essential device, and might bring more known requirements from the domain; however, for this solution, it is not necessary, given the non-sensitive level of information processed, and non-critical business need within the environment.

The Scrum Alliance agile framework organizes its work into stages called "sprints", each designed to work on either discovery in support of business goals (such as getting estimates for building complex feature sets) or to deliver software in a step-by-step, measured manner. Sprints are time-bound, typically 1 to 3-week periods, in agile methodology, where developers work against written requirements known as "user stories". Each user story specifies for the software development team: a known user or stakeholder role for which the work will be delivered (i.e.: "an elementary school teacher", "a school board", "a group of new parents to the school district", "a preschool student" and so on), a brief description of the work ("the student will login using a QR code and then be forwarded to with an academic progress dashboard upon presenting successful pre-existing printed credentials") and a short paragraph of the desired outcome ("so they can view their scores before going to an activity task."). The user story is intended to provide enough detail for stakeholders to understand the work at a high level, yet also enough clarity and specification for the technical teams to deliver on its outputs, outcomes, and end results. This proposal will use 1-week sprints and plan for 3 sprints' worth of work to execute the overall project. Each sprint will represent 35 hours of work for one senior developer to execute, using AI-assisted code generation for connection technology components and for infrastructure deployments and compliance checks. In a world of 2026, where AI is now a mainstay in the workplace, this scenario emulates what a real-world task team might do for a senior technologist working in a K-12 education technology product company, and a prototype they might develop to test a concept. The project will create user stories, work on each user story in the sprint, and drive toward desired results and outcomes for a secure, low-cost messaging solution.

Sprint 1

- User Story 1
    - Title: Gather solution requirements and final details for solution plan

[4] Scrum Alliance. (n.d.). *Scrum Alliance: Transforming the world of work*. Retrieved January 28, 2026, from https://www.scrumalliance.org

    - Details: Gather requirements for the solution within a K-12 context for domain-fit, use cases, security, and other needs to deliver
    - Outcome: Listing of business and functional requirements for a custom solution in the K-12 environment using open/low-cost technology
- User Story 2
    - Title: Research components of the solution plan and model in the working architecture
    - Details: Find end-to-end solution using open-source/broadly available platforms & technology paired with Raspberry Pi hardware to make a secure, reliable delivery system for K-12 environments
    - Outcome: Listing of technologies and strategy to wire up for the solution
- User Story 3
    - Title: Deliver final working solution proposal with technical architecture and work breakdown
    - Details: Deliver a technical brief and an executive summary of the overall solution
    - Outcome: Have an approval game plan before moving to the next sprint with a technical solution and stakeholder signoff

Sprint 2

- User Story 4
    - Title: Prepare backend delivery solution component
    - Details: Consider Google Spreadsheets with 2FA support to gather data securely with API support for the backend that would be found within an enterprise, or another tech solution that suits the K-12 agency's overall requirements
    - Outcome: Have a messaging backend that works in K-12 agencies
- User Story 5
    - Title: Prepare frontend delivery solution component
    - Details: Consider Raspberry Pi or similar low-cost hardware to display via HDMI to a school display from the content from UC4
    - Outcome: Have a messaging frontend that works in K-12 agencies
- User Story 6
    - Title: Integrate a complete solution
    - Details: Bring together all pieces for an end-to-end solution
    - Outcome: Connect and prove out the complete solution; report out before Phase 3

Sprint 3

- User Story 7
    - Title: QA Test and Package Solution
    - Details: Prepare product for qa testing and emulate go-to-market

    - Outcome: Work out bugs and fix the final solution
- User Story 8
    - Title: Final Package Solution w/ Demo Video
    - Details: Wrap up the product with a demonstration video to showcase
    - Outcome: Present and show how to use the final solution
- User Story 9
    - Title: Close down and report out
    - Details: Shut down the final project and wrap-up from PoC
    - Outcome: Gather all data from the milestone and report out on the findings

A. 3. A Justification of the Plan

Agile methodology is widely used today in many software development shops and in K-12 environments as well. As seen in guidance from the Computer Science Teachers Association[5], Agile methods "not only mirror the real world of computing, but also foster essential 21st-century skills: communication, adaptability, and creative problem-solving." Is it designed to be adaptive to changing requirements, as a dynamic learning environment can be? Numerous open-source communities, such as Moodle, Apereo, Sakai, Ed-Fi, and others, use agile as a basis for developing their technology offerings, with success in K-12 and higher education environments as well.

Agile is feasible, especially for a proof-of-concept or prototype project without clear goals or an end state in mind. Agile lends well, as the sprint structure breaks work into manageable chunks, allowing stakeholders to provide input at each stage to monitor and shepherd the project within its timelines and budget. Agile methodology makes it acceptable to "fail fast" and course-correct quickly week to week. In contrast, other models may slow progress depending on gating conditions or require more cycles for verification. Given that the information processed in the proposal is neither sensitive nor considered personally identifiable information, and in line with FEPRA guidance, a quick, adaptive execution model will suit the outcomes of this project well. If the project is not feasible as imagined, we can report progress as an experiment and work with stakeholders to determine the next steps for the project phase.

B. Review of Other Work and (B1) Works Informing Design

To support this proposal, four primary sources of reference will be utilized to inform its design and execution:

[5] Quinn, C. (2026, January 21). *Iterate to innovate: Using Agile methods to build student engagement*. Computer Science Teachers Association. https://csteachers.org/iterate-to-innovate-using-agile-methods-to-build-student-engagement/

- "Raspberry Pi in Education"[6]from the Institute of Electrical and Electronics Engineers (Balon, T., & Simić, M., 2019), an industry survey on how the low-cost Raspberry Pi computing device is used in high school and higher education settings. This work will provide a reference to why the Raspberry Pi came into existence, the problems it attempts to solve, and how it functions today in educational environments.
- "FIPS Publication 140-2"[7], from the National Institute of Standards and Technology (2001), is a federal government specification and reference book on various encryption techniques and standards. This work will be used to select the appropriate methods and technologies to support the proposal design for an end-to-end secure solution, using technologies aligned with the standards outlined and recommended here.
- "The K12 SIX Essential Series"[8], from The K12 SIX (n.d.), a non-profit and industry leader in K-12 cybersecurity topics and convenings, comprising K-12 school districts, experts, and industry leaders. This work will provide the framework and context for working in K-12 with a cybersecurity focus, guiding the solution components and connections.
- "Delivering on the Promise of Digital Equity"[9], from Digital Promise (Weaver, D. J., 2022), a leading nonprofit that researches and advises on K-12 digital access and equity as students and teachers use technology to improve education outcomes and achievement. This work will help address aspects such as the need for low-cost computing, the benefits of connected environments, and the overall technology safety principles when using digital tools to improve education outcomes and equity.

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[6] Balon, T., & Simić, M. (2019). Using Raspberry Pi computers in education. *2019 42nd International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*, 671–676. https://doi.org/10.23919/MIPRO.2019.8756967

[7] National Institute of Standards and Technology. (2001). *Security requirements for cryptographic modules* (FIPS Publication 140-2). U.S. Department of Commerce. https://doi.org/10.6028/NIST.FIPS.140-2

[8] K12 Security Information eXchange. (n.d.). *The essentials*. K12 SIX. Retrieved January 28, 2026, from https://www.k12six.org/essentials-series

[9] Weaver, D. J. (2022, December). *Delivering on the promise of digital equity*. Digital Promise. https://doi.org/10.51388/20.500.12265/166

D. Current Project Environment

The proposal solution is designed to align with K-12 educational organizations and culture in the following ways:

- The solution is developed with the budgetary limitations and technology access a K-12 education organization is likely to have, such as Raspberry Pi computers and HDMI-enabled TV and projector devices. The solution does not require new contracts or expensive purchases, which are delayed in a procurement cycle and/or require additional support to put in place.
- This solution is designed to deliver a modern, secure solution that meets the actual needs of schools and districts, as seen in many educational organizations that need to distribute organizational information and announcements. It will use modern technology that has typically passed security reviews and operational checks, and provide a solution that is not breakable by casual hackers or standard attack tools. Given its context, it is considered a moderate-risk solution, so it will be secured accordingly and with the information it processes, which helps keep the overall solution cost down and applies to the low-cost needs mentioned above.
- The solution is also cost-efficient and does not depend on additional resources, as printing or posters may, and is considered a "green" solution. The solution is designed to replicate real-time information from school systems that schools are used to, in a way that is relevant to how students receive information in their day-to-day settings.
- The solution can be administered by IT staff in school settings, by those acting in that role, or potentially by trusted senior students, such as "tech crew" or similar afterschool groups.
- The solution will provide a strategy for up-to-date information, with a solution that is secure and unbreakable by standard methods, and deliver a fraction of the cost of commercially procured solutions. It will have commercial-grade protections at a commodity-level cost to make it most effective for public-service agencies. With a usable, practical solution, a school will get its information quicker out to more students, so they can be equipped to perform at their best within their pathways and goals.

E. Methodology

The software methodology used for this project will be the Scrum Alliance "agile" framework. This framework offers a flexible, proof-of-concept approach that enables discovery and quick proof-of-concept execution to determine whether this concept will work. Given this is an informational product, and not a business-critical or life-sustaining system, it can move quickly through its work cycles, or sprints in agile methodology, and "fail fast" by design to expose the pieces that are not working to the stakeholders as it happens. Also, the agile method will deliver outcomes at each stage and produce measurable feedback and data, so the activity is valuable regardless of what happens in the end, to inform the overall development of the following stages.

F. Project Goals, Objectives, and Deliverables

F1. Goals, Objectives, and Deliverables Table

Goal	Supporting Objectives	Deliverables Enabling the Project Objectives
Prove open & low-cost tech can provide commercial-grade secure solutions	Can a commonly available school platform, such as Google Spreadsheets, serve as a secure backend?	Prepare backend delivery solution component (Google Spreadsheet w/ APIs)
		Deliver the final working solution proposal, including the technical architecture and work breakdown.
	Can a commonly available PC device (such as a Raspberry Pi) run a secure video solution?	Prepare frontend delivery solution component (Raspberry Pi)
		Deliver the final working solution proposal, including the technical architecture and work breakdown.
	Can the solution work end-to-end, and how can we have information assurance?	Integrate a complete solution.
		QA test
		Demo video

F2. Goals, Objectives, and Deliverables Descriptions

- Goal: Prove open & low-cost technology can provide commercial-grade, secure solutions - Typically in K-12 environments, a blocker to adoption can be cost, procurement, or sub-standard solutions which do not meet organizational requirements. Sometimes, however, in academic environments, open-technology solutions do find a way to provide enterprise-grade solutions at a free or low cost. This project attempts to prove a concept along those lines, where commonly available technology can be used to provide an enterprise-grade solution to benefit public agencies.
- Objective 1: Can a commonly available school platform, like Google Spreadsheets, be a secure backend? - Typically, a blocker to adoption in K-12 environments can be complex data integration or expensive products that take months and years to implement. This project will test a platform such as Google Spreadsheets, a Google for Education product and a low-cost option found in schools, to see if it can serve as an everyday driver backend for this system. With this, the data will be secure and protected by modern measures, while providing a lightweight,

maintainable system that the usual office staff can use without additional training or certification.

    - Deliverable 1: Prepare backend delivery solution component (Google Spreadsheet w/ APIs) - this will utilize Google Spreadsheet with their secure API backend for data delivery per authenticated user, as an accessible and available sub-system.
    - Deliverable 2: Deliver the final working solution proposal with technical architecture and work breakdown. The selected backend for the project will be integrated into the complete solution, with documentation and technical architecture on how it works, along with a proof-of-concept delivery work breakdown.
- Objective 2: Can a commonly available PC device (Raspberry Pi) drive a secure video solution? -Typically, a blocker to adoption in K-12 environments can be expensive hardware devices. This objective will test whether a low-cost hardware device, such as a $10-30 Raspberry Pi, can provide similar functionality, and whether it is likely to be found in computer or science labs.
    - Deliverable 1: Prepare frontend delivery solution component (Raspberry Pi) - this will utilize a Raspberry Pi with HDMI video output, USB power, and WiFi network connections to provide the video aspect of the solution
    - Deliverable 2: Deliver final working solution proposal with technical architecture and work breakdown - this will be a part of the overall final working solution, which fits the defined technical architecture, and with a work-breakdown schedule estimate to deliver it
- Objective 3: Can the solution work end-to-end, and how can it ensure information assurance? - Once the first two pieces are put together, can the whole thing operate as a complete solution?
    - Deliverable 1: Integrate complete solution - bring all pieces together as an end-to-end solution.
    - Deliverable 2: Quality Assurance testing - ensure the end-to-end solution works by using comprehensive testing to verify that the outcomes and end goals are met.
    - Deliverable 3: Demo video - demonstrate the concept with a final video of how the product works, emulating a real-life product video of the solution.

G. Project Timeline with Milestones

Milestone	Duration (hours or days)	Projected Start Date	Anticipated End Date
Sprint 1- Research & Design	5 days	02/09/2026	02/13/2026
Sprint 2 - Implementation	5 days	02/16/2026	02/20/2026
Sprint 3 - QA, Demo & Wrap-Up	5 days	02/23/2026	02/27/2026

H. Outcome

The desired outcome of this project and proposal is to develop a proof-of-concept type system that provides secure end-to-end message delivery within a K-12 school context. So a school administrator would enter information in a secure data entry screen, such as a private Google Spreadsheet, and the solution would deliver the message securely to a Raspberry Pi device, to display on a TV or projector, such as in the setting of an entry way or cafeteria for students and staff to see. This project is set to be delivered within a 3-week time frame, using the agile software methodology, and using a work-week schedule to deliver each phase. Data will be collected through technical research, domain research, technical designs, and results from end-to-end tests. Data will be measured on how well we executed each project sprint. How well did we deliver on business requirements for secure delivery at a low cost? What was missed and estimated costs for impact, if so? If successful, how long is each message in a cycle, and how long should a user be expected to wait?

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